```
import matplotlib.pyplot as plt  #Pragya singhal, CU24220050, 43
import numpy as np
import pandas as pd
import seaborn as sns
# Set visual style for charts
sns.set_theme(style="whitegrid")
print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
# Load dataset directly from session storage #Pragya singhal, CU24220050, 43
filename = "Global Superstore.csv"
try:
  df = pd.read_csv(filename, encoding="latin1")
  print(f"Dataset '{filename}' successfully loaded!")
except FileNotFoundError:
  print(
      f"Error: '{filename}' "
      " not upload."
  )
display(df.head())
```

```
Dataset 'Global Superstore.csv' successfully loaded!
```

1 to 5 of 5 entries Filter

index	Category	City	Country	Customer ID	Customer Name	Discount	Market	ji_lu-shu	Order
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders	0.0	US	1	00:00.
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff	0.0	US	1	00:00.
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	00:00.
3	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	00:00.
4	Office Supplies	Los Angeles	United States	AP-109154	Arthur Prichep	0.0	US	1	00:00.

Show 25 per page
Like what you see? Visit the data table notebook to learn more about interactive tables.

```
Warning: Total number of columns (27) exceeds max_columns (20) limiting to first (20)
```

```
# Convert 'Order Date' to datetime format   #Pragya singhal, CU24220050, 43
df["Order Date"] = pd.to_datetime(df["Order Date"], errors="coerce")

# Extract Year and Month for trend analysis
df["Year"] = df["Order Date"].dt.year
df["Month-Year"] = df["Order Date"].dt.to_period("M")
```

```
print("Data cleaning & preprocessing completed successfully!")
```

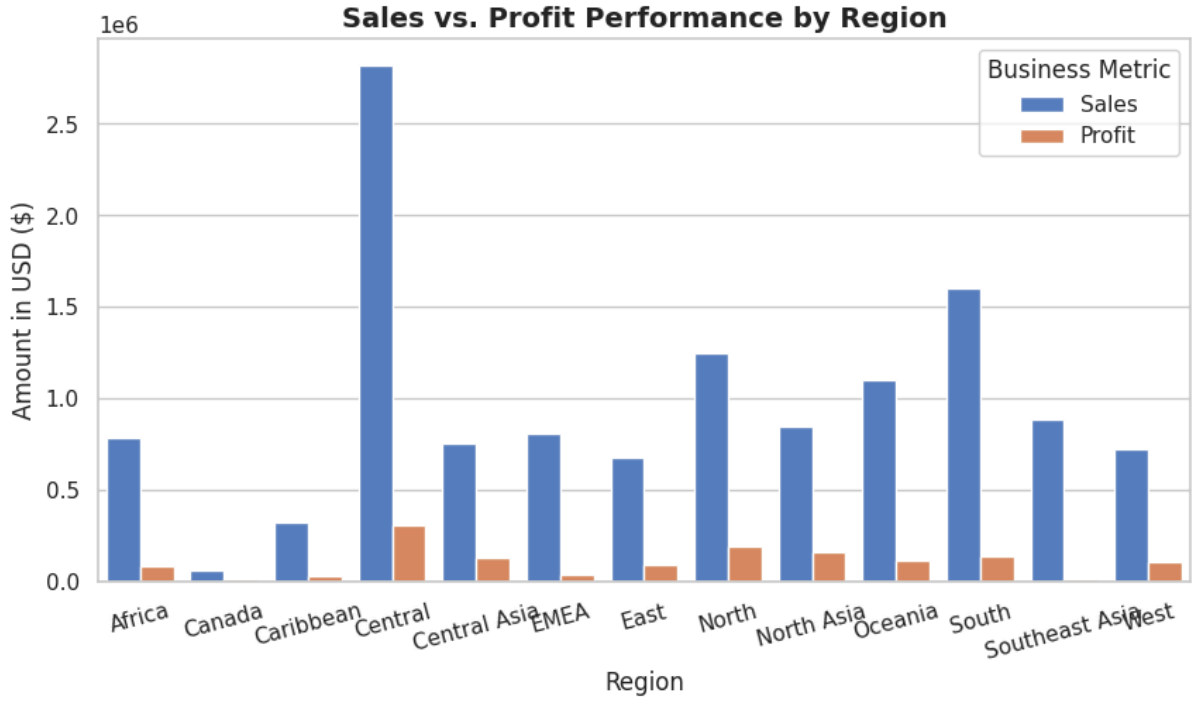
```
Data cleaning & preprocessing completed successfully!
```

```
total_sales = df["Sales"].sum()   #Pragya singhal, CU24220050, 43
total_profit = df["Profit"].sum()
profit_margin = (total_profit / total_sales) * 100
total_orders = df["Order ID"].nunique()
total_customers = df["Customer ID"].nunique()

print("========================================")
print("       KEY PERFORMANCE INDICATORS (KPIs)  ")
print("========================================")
print(f" Total Revenue (Sales) : ${total_sales:,.2f}")
print(f" Total Net Profit      : ${total_profit:,.2f}")
print(f" Overall Profit Margin : {profit_margin:.2f}%")
print(f" Total Unique Orders   : {total_orders:,}")
print(f" Total Unique Customers: {total_customers:,}")
print("========================================")
```

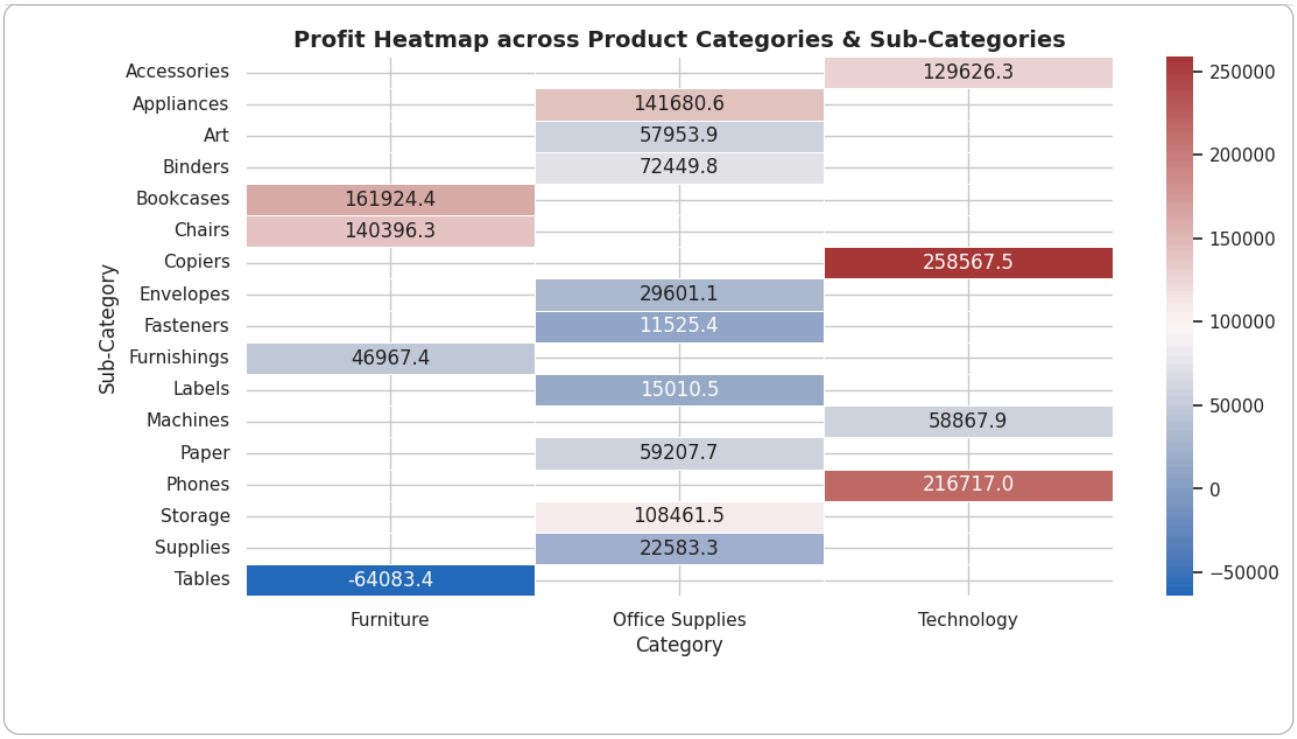
```
========================================
       KEY PERFORMANCE INDICATORS (KPIs)
========================================
 Total Revenue (Sales) : $12,642,905.00
 Total Net Profit      : $1,467,457.29
 Overall Profit Margin : 11.61%
 Total Unique Orders   : 25,035
 Total Unique Customers: 4,873
========================================
```

```
plt.figure(figsize=(10, 5))   #Pragya singhal, CU24220050, 43
regional_df = (
    df.groupby("Region")[["Sales", "Profit"]].sum().reset_index()
)
regional_melted = regional_df.melt(
    id_vars="Region",
    value_vars=["Sales", "Profit"],
    var_name="Metric",
    value_name="Amount",
)
sns.barplot(
    data=regional_melted,
    x="Region",
    y="Amount",
    hue="Metric",
    palette="muted",
)
plt.title(
    "Sales vs. Profit Performance by Region", fontsize=14, fontweight="bold"
)
plt.ylabel("Amount in USD ($)")
plt.xlabel("Region")
plt.xticks(rotation=15)
plt.legend(title="Business Metric")
plt.show()
```




```
plt.figure(figsize=(12, 6))   #Pragya singhal, CU24220050, 43
pivot_table = pd.pivot_table(
    df,
    index="Sub-Category",
    columns="Category",
    values="Profit",
    aggfunc="sum",
)

sns.heatmap(
    pivot_table,
    annot=True,
    fmt=".1f",
    cmap="vlag",
    cbar=True,
    linewidths=0.5,
)
plt.title(
    "Profit Heatmap across Product Categories & Sub-Categories",
    fontsize=14,
    fontweight="bold",
)
plt.xlabel("Category")
plt.ylabel("Sub-Category")
plt.show()
```




```
plt.figure(figsize=(14, 5))   #Pragya singhal, CU24220050, 43
monthly_sales = df.groupby("Month-Year")["Sales"].sum()
monthly_sales.index = monthly_sales.index.to_timestamp()

plt.plot(
    monthly_sales.index,
    monthly_sales.values,
    marker="o",
    color="teal",
    linewidth=2,
)
plt.title("Overall Monthly Sales Trend Over Time", fontsize=14, fontweight="bold")
plt.xlabel("Order Month")
plt.ylabel("Total Sales ($)")
plt.grid(True, linestyle="--", alpha=0.7)
plt.show()
```

